Where Should the Gradient Go?

Can an oracle-text diagnostic predict which side of a small VLM to adapt?

CS 8803-VLM, Fall 2026 · Project Proposal · Team: [names]

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1. Problem statement

When adapting a small vision-language model to a visual question-answering task, a practitioner must decide where to put a limited fine-tuning budget. One practical baseline is to freeze the visual components and apply LoRA to the language model. Another is to adapt the vision encoder and its connector while keeping the language model frozen. We ask whether a diagnostic measured before fine-tuning can predict which placement will work better for a particular task-model pair.

Our core study uses two models, Qwen3-VL-2B-Instruct (Bai et al., 2025) and InternVL3-2B (Zhu et al., 2025), on two tasks, ChartQA and Geometry3K. These four task-model pairs are the units of evidence for the prediction claim. Each pair receives two placements and three training seeds, for **24 main training runs**.

Input. A pretrained model, a target task's training data, a separate labeled diagnostic set with audited visual descriptions, and an adaptation budget defined by trainable parameters and optimizer steps.

Output. A diagnostic profile for each pair, a placement prediction recorded before seeing fine-tuning outcomes, and checkpoints and evaluation results for both placements across all three seeds.

H1. Pairs with a larger fraction of errors rescued by replacing the image with a faithful textual representation will show a larger accuracy advantage for vision-side adaptation over language-side adaptation. We will also test a simple, fixed rule that predicts which placement wins from this fraction.

This is a hypothesis about the usefulness of a diagnostic, not an assumption that visual errors originate exclusively in the vision encoder. A language model can fail to use otherwise adequate visual features, and language-side adaptation might repair that failure.

Whether oracle-text rescue helps choose an adaptation location is precisely what the experiment tests.

Success criterion. The diagnostic should help select placements across pairs and reduce average selection regret relative to always choosing either placement. We will report the direction and uncertainty of each placement difference, prediction outcomes, and failures. If one placement wins everywhere, a fixed recipe may be sufficient for this sample. Four pairs can support an initial test, but cannot establish broad generalization to unseen tasks and model families.

2. Motivation

Compact VLMs make multimodal inference feasible under limited memory and latency budgets (Marafioti et al., 2025; Huang et al., 2026). Their downstream adaptation also needs to be affordable. Trying many adapter configurations can cost more than the final fine-tuning run, so a useful diagnostic would help decide where to spend that budget before a full search.

Chart and geometry questions are useful test cases because they require both extracting visual information and using it to answer a question. A wrong answer alone does not tell us which step failed. Supplying a textual rendering of the visual content lets us measure how the model behaves when some of the visual processing burden is removed. However, this intervention also changes the input representation, so we treat it as a predictive feature rather than a definitive attribution of the error to a particular module.

The immediate beneficiaries are practitioners adapting small VLMs for chart and diagram understanding. Our course project focuses on these tasks rather than claiming transfer to unrelated application domains.

3. Related work and the specific gap

Diagnosing visual dependence. MathVerse varies the information available in text and images across versions of visual mathematics problems (Zhang et al., 2024). CapGeo-Bench studies geometric perception and reports improved reasoning performance when models receive high-quality captions (Li et al., 2025). These works motivate our image-to-text substitution diagnostic; we do not claim to invent that intervention. CVS studies whether adding a question changes a model's assessment of an answer given an image,

and uses the resulting signal for data selection (Sun et al., 2026). Its conditioning intervention differs from our question-only control.

Adapter placement and visual fine-tuning. LoRA provides the adaptation mechanism (Hu et al., 2021). PLoP already selects adapter module types using information from a pretrained model and a target task (Hayou et al., 2025). Image-LoRA selects attention heads with a gradient-based probe and restricts adapter updates to visual-token positions (Luo et al., 2025). Thus, diagnostic-guided adapter selection is already an established direction. Our question is narrower: can an inference-only oracle-text signal predict a choice between the visual pathway and the language decoder?

CoVFT is a particularly relevant comparison. Zhou et al. (2026) find that several visual fine-tuning methods improve average performance but do not improve every task. They also introduce a context-aware method that makes visual fine-tuning more consistent. Their setup trains the language model and projector while varying visual adaptation; our experiment instead allocates a matched adapter budget to one side or the other. CoVFT therefore motivates examining task-dependent outcomes, but does not directly answer our placement question.

Perception-aware post-training. PAPO adds perception-aware optimization to multimodal reinforcement learning (Wang et al., 2025). Wu et al. (2026) analyze an imbalance between perception and reasoning improvements during post-training and investigate changes to supervision. These studies motivate examining perception, but our experiment keeps the answer-only SFT objective fixed and changes only the adaptation configuration.

Compact models and evaluation. SmolVLM and the efficiency analysis of Huang et al. study compact multimodal systems (Marafioti et al., 2025; Huang et al., 2026). We use the Qwen3-VL and InternVL3 families to test whether a diagnostic behaves similarly across two architectures. ChartQA and Geometry3K provide the core tasks and structured annotations (Masry et al., 2022; Lu et al., 2021). CharXiv and ChartQAPro provide optional external chart evaluations (Wang et al., 2024; Masry et al., 2025).

Our contribution. We propose a controlled test of the link between an oracle-text diagnostic and a binary adaptation decision in small VLMs. The contribution would be evidence about when this signal succeeds or fails, together with reproducible diagnostics, matched-budget comparisons, and explicit limits on the prediction claim.

4. Preliminary approach

Stage A: Diagnostic before fine-tuning

We separate adaptation data, diagnostic data, and final evaluation data. The diagnostic set comes from validation data or a fixed partition of the training split that is excluded from adaptation. Splits are grouped by source chart or geometry problem so that related questions do not cross partitions. Final test labels are not used to build the diagnostic, choose a placement, tune a threshold, or adjust prompts.

For each diagnostic item, we evaluate the pretrained model under three conditions using fixed prompts and deterministic decoding:

Condition	Input	Purpose
Image	Image and original question	Measure ordinary task performance
Oracle text	Audited visual description and original question, without the image	Measure errors rescued by a textual representation
Question only	Original question without the image or visual description	Measure performance available without visual input

For questions with answer options, the options are identical across conditions. Among items answered incorrectly in the image condition, we first separate those answered correctly with the question alone. Among the remaining errors, we distinguish **oracle-rescued errors** from **residual errors**, depending on whether oracle text produces a correct answer. These categories are operational measurements. Question-only correctness can reflect guessing or memorization, and a residual error can reflect reasoning, text interpretation, or an incomplete oracle.

The primary diagnostic is

[p_{t,m}=\frac{N_{\text{oracle-rescued}}}{N_{\text{oracle-rescued}}+N_{\text{residual}}}.]

We also report the category counts, their fractions among image-condition errors, oracle-rescued errors as a fraction of all diagnostic items, and question-only accuracy on the full diagnostic set. This makes the denominator explicit and distinguishes a high rescue

fraction from a large absolute number of recoverable errors. Image-correct items that fail with oracle text are reported separately as a check on the intervention.

Oracle construction. ChartQA's original release includes underlying data tables. We will serialize the complete relevant chart table using a fixed, question-independent format rather than select the row that answers a question. A table alone may omit colors, spatial relations, or other visual encodings, and may contain greater numerical precision than the displayed chart. We will audit completeness and precision, add visible metadata where needed, and report the coverage of usable items. Geometry3K provides annotated diagram and text logic forms. We will use annotations of given facts, rendered in a fixed readable format, and exclude solutions, theorem sequences, and derived target values.

We will manually inspect 100 diagnostic examples per task for completeness, answer leakage, and representation changes. Question-only evaluation measures nonvisual answerability; it does not detect leakage introduced by oracle text. Items lacking a faithful oracle will be excluded from the primary diagnostic using documented criteria, with exclusion counts and coverage reported. Model-generated descriptions will not be silently substituted for ground-truth annotations.

Stage B: Two placements, three seeds

All main runs use the same image-question-answer training examples and answer-only SFT. Oracle descriptions are used for diagnosis only.

Placement	Adapted components	Frozen components
Language-side LoRA	Language decoder linear layers selected in advance	Vision encoder and visual connectors
Vision-side LoRA	Vision encoder linear layers and visual connector linear layers selected in advance	Language decoder

Here, the visual connector means the model-specific merger or projector; all relevant connector paths will be enumerated explicitly, including Qwen3-VL's additional visual feature integration paths. We compare the complete vision-side configuration with the language-side configuration. Without a separate connector ablation, we cannot attribute an observed benefit specifically to the encoder.

For each model, the module lists, ranks, and LoRA scaling will be fixed before inspecting task evaluation outcomes. Ranks will be chosen to match trainable-parameter counts within 10%, with the exact counts reported. Paired runs use the same three seed IDs, training examples and order, effective batch size, optimizer steps, answer targets, and evaluation settings. Image preprocessing is fixed within each model and shared by both placements.

Matching parameters and steps does not guarantee equal FLOPs, GPU time, or peak memory. We will measure those quantities separately. In particular, freezing the decoder's weights does not remove the need to backpropagate through it to train visual adapters. Results will be described as a comparison under a fixed training recipe, not as proof that either placement is optimal after unrestricted tuning.

The committed grid is:

Task	Model	Placements	Seeds per placement	Main runs
ChartQA, human-authored subset	Qwen3-VL-2B-Instruct	2	3	6
ChartQA, human-authored subset	InternVL3-2B	2	3	6
Geometry3K	Qwen3-VL-2B-Instruct	2	3	6
Geometry3K	InternVL3-2B	2	3	6
Total	**4 task-model pairs**			**24**

Both placements are evaluated on all four pairs. Zero-shot evaluation adds no training runs. Optional Qwen3-VL-8B-Instruct inference provides a larger-model reference, not an accuracy ceiling or another adapted pair. Merger-only adaptation, split-budget adaptation, auxiliary supervision, Image-LoRA training, and random-placement training are outside the committed grid.

Stage C: Prediction and evaluation

For each pair, define the placement difference in percentage points as

[\Delta_{t,m}=\frac{1}{3}\sum_{s=1}^{3}\left(A^{\mathrm{vision}}*{t,m,s}-A^{\mathrm{language}}*{t,m,s}\right).]

The primary analysis plots this difference against the diagnostic rescue fraction and reports all four observations. For a concrete prediction test, we preregister a simple rule: choose vision-side adaptation when more than half of the non-question-only image errors are rescued by oracle text, and choose language-side adaptation otherwise. The 0.5 threshold expresses a majority-rescue heuristic; it is not an established optimal threshold. If the diagnostic denominator is zero, we report that the rule is undefined for that pair and use language-side adaptation as an operational fallback, without counting it as a diagnostic prediction.

This rule is recorded before any placement outcomes are inspected and is not fitted to the four pairs. We report its predictions, empirical win count, and mean selection regret, defined as the observed accuracy gap between its chosen placement and the better of the two placements on each pair. We compare it with always-language and always-vision policies using the same results. A disagreement between a useful continuous trend and a poor threshold rule will be reported rather than repaired by tuning the threshold on these pairs.

The four pairs are a small crossed sample with shared tasks and model families. Neither the three seeds nor the number of evaluated questions increases the number of task-model pairs. Correlations and policy regret are descriptive at this scale. We will not claim a learned selector validated on held-out tasks and models; that would require a larger, separately planned pair-level study.

Metrics and uncertainty. ChartQA uses its official relaxed accuracy; Geometry3K uses answer-option accuracy with fixed parsing. Any optional external evaluation uses that benchmark's own protocol. We report all three seed results, the mean paired difference, and its seed variability. We also compute paired bootstrap intervals over source charts or geometry problems, preserving the pairing of placements and all questions from a source. These intervals describe evaluation-sample uncertainty conditional on the trained seeds. A

paired t interval over the three seed-level differences is reported separately to show training uncertainty, with its strong small-sample limitation. A narrow question-level interval alone is not evidence of a stable training effect.

Three seeds remain the fixed design. Earlier team runs required four seeds to establish an approximately two-point effect, so three should not be presented as guaranteed sufficient power for smaller placement differences. We will distinguish a detectable difference, a result too imprecise to resolve, and evidence against the diagnostic. An interval containing zero does not demonstrate equivalence, and a failure to resolve the effect will not be reported as successful prediction.

Feasibility, optional evaluations, and attribution

We will profile both placements on the available PACE GPU before launching the grid, recording peak memory, step time, effective batch size, resolution, and visual-token count. The initial plan uses a 768-pixel longer-side cap, subject to a readability check on charts and diagrams. Any memory-driven restriction of visual layers or preprocessing will be finalized before the main runs and applied consistently across the two tasks for that model. Short profiling runs and any failed or repeated jobs will be disclosed separately from the 24 main runs.

If resources remain after completing the core grid, we will evaluate the chart-adapted checkpoints on CharXiv and ChartQAPro without additional fine-tuning. These are distribution-shift checks, not additional independent adaptation pairs. Descriptive questions in CharXiv offer a complementary measure of chart reading, but do not supply a complete oracle for every reasoning question.

The training and evaluation harness is reused from a team member's prior project. Its repository and commit will be identified in the final code packet, together with the new diagnostic and placement code. AI-assistant contributions will be recorded in AGENT_USAGE.md. Main results include every planned seed; any representative checkpoint will use a seed chosen in advance rather than the best test seed.

5. Timeline and decision criteria

Dates	Deliverable

Sep 22–26	Finalize source-grouped splits; build oracle serializers; audit 100 diagnostic items per task; profile both placements
Sep 29–Oct 3	Complete diagnostic profiles for both models on both tasks; freeze module lists, ranks, training settings, and all four placement predictions
Oct 6–10	Launch the complete 24-run grid; verify training and evaluation outputs; mark incomplete seed sets as preliminary
Oct 13–15	Midterm report: all four diagnostic profiles, completed paired comparisons with three seeds wherever available, and explicit status of unfinished runs
Oct 16–Nov 10	Finish all 24 main runs and uncertainty analyses; conduct optional external chart evaluations only after the core grid is complete
Nov 17	Code and audit packet with configurations, all seeds, predictions recorded before training, exclusions, and compute logs
Nov 24	Rebuttal and final revisions
Dec 1/3/8	Final presentation, according to the assigned course slot

Midterm checks. We will assess whether the oracles are faithful and sufficiently representative, whether the diagnostic varies across pairs, and whether both placements are feasible under the declared budget. Incomplete one- or two-seed results will not establish a winner or trigger a scientific null claim. We will finish the committed comparisons where technically feasible before deciding whether expansion is warranted.

Final interpretation. The study can produce evidence that the diagnostic is useful, evidence that it predicts the wrong placement, or an inconclusive result because the sample or effects are too small. If all profiles are similar, the sample cannot test much predictive variation. If one placement consistently wins, the conditional selector has not demonstrated a benefit over that fixed choice. If uncertainty dominates, we will report the minimum effects the study can resolve and avoid claiming that vision-side adaptation is ineffective in general. A null result in this setup is not a direct replication of CoVFT.

6. References

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